Privacy For Sale

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October 2026

Motivation

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- ∼ Firms collect consumer data as by-product of ordinary transactions
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This raises welfare concerns

- ∼ Firms may profit from selling data even when its use harms consumers
- ∼ And consumers may forgo purchases they would otherwise value

Motivation: Data Bans

A natural policy response is to ban third-party data sales $\rightsquigarrow$ **Data Ban**

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- $\sim$ If consumers dislike having their data sold and used, a data ban removes the harm directly

However, if selling data is valuable, seller may reduce product prices to acquire more data

- $\sim$ Thus, a data ban may harm consumers by making products more expensive

Motivation: Data Markets

Another policy response is to give consumers more control over their data:

- ∼ How? Enable firms to credibly promise not to sell consumer data
- ∼ Firms could then offer different versions of the same product, differing in whether/how data will be monetized
- ∼ Different product prices will implicitly compensate consumers for their data ⇝ **Data Market**

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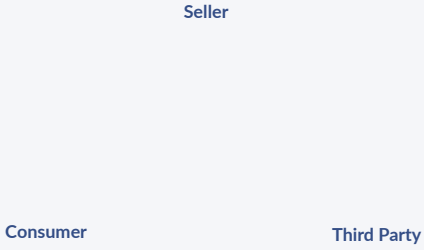
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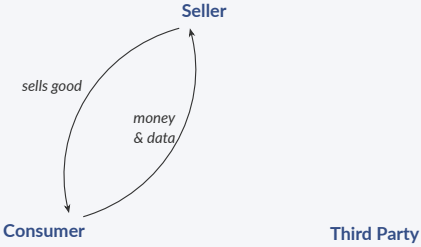
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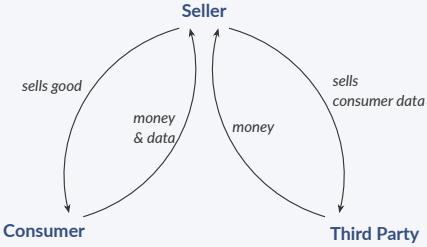
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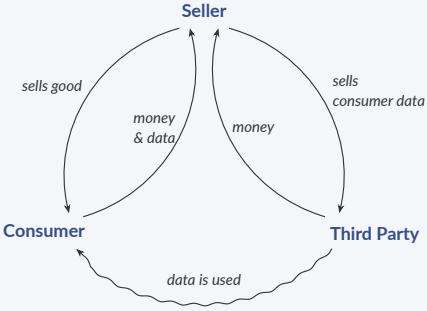
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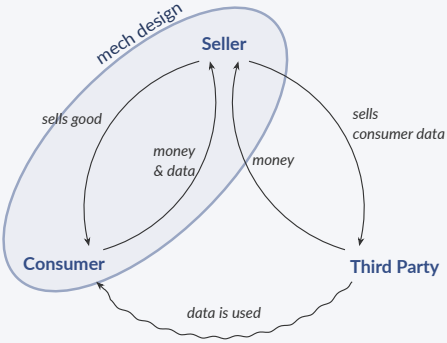
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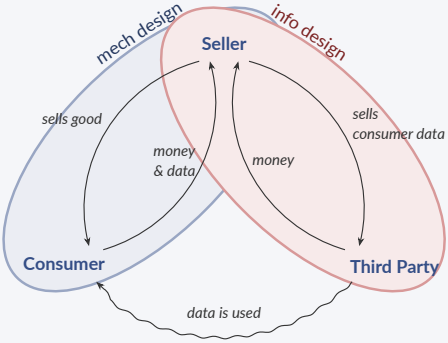
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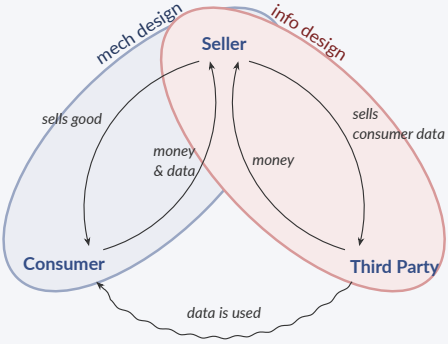
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Tradeoff:

Mechanism shapes data sale;

data sale shapes mechanism

This Paper

1. A tractable framework to study third-party data sharing

- ∼ It draws on mech and info design to jointly determine product prices, data disclosure, and downstream data use

2. Use this framework to study the welfare effects of regulation

- ∼ Anarchy, Ban, or Market?

Preview of Results

A key parameter shapes equilibrium and policy:

- $\sim$ The **privacy** $\beta \rightsquigarrow$ It governs whether consumers with higher WTP for seller's good have higher ($\beta > 0$) or lower ($\beta < 0$) WTP for privacy

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In particular,

- If $\beta > 0$, eqm data sharing is **coarse** and more data use **hurts** consumers
- If $\beta \leq 0$, data sharing is **granular** and more data use **benefits** consumers

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Without further structure, each regime can be best for all consumers

- ∼ We derive structural conditions that resolve these welfare comparisons
- ∼ **Data market** is robust compromise with many desirable welfare properties

Related Literature

We contribute to growing literature on the **economics of data and privacy**

Acquisti, Taylor, Wagman '16, Bergemann, Bonatti, Smolin '18, Bergemann and Bonatti '19, Goldfarb, Tucker '23, Galperti, Levkun, Perego '24, Baley and Veldkamp '25, Galperti, Liu, Perego '26, Ichihashi, Jeon, Kim (2025), Ali, Lewis and Vasserman (2023) [and many others]

In particular, Bisceglia, Bonatti, Scott-Morton (2025)

We contribute to literature on **mech design with info externalities**

Calzolari and Pavan '06, Rayo '13, Dworczak '20, Dworczak '21, Krahmer, Strausz '23, Rivera Mora '25, [and many others]

In particular, Doval, Skreta '26

model

Model: First Period

The seller has a single indivisible good for sale

$\theta \in [0, 1]$ is the consumer's valuation for this good, distributed with CDF F

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Seller announces a (deterministic) **mechanism** $(x, t) : M \to \{0, 1\} \times \mathbb{R}$

- $\rightsquigarrow$ If $x(m) = 0$, product is not allocated, game ends, payoffs are zero
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In the latter case, we refer to message m as the consumer's **data record**

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- $\sim$ we interpret action a as **data use** — e.g., intensity of third-party advertising

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Note: seller controls whether data are sold, but cannot contract on their use

Data Policies

We compare three alternative policy environments:

- **Data Anarchy**. The seller **cannot commit** to the second-period data-pricing strategy $p : M \to \mathbb{R}$
- **Data Market**. The seller **can commit** to the second-period data-pricing strategy $p : M \to \mathbb{R}$
- **Data Ban**. The seller **cannot sell** consumer data to the third party

Payoffs: Seller

The **seller**'s payoff is

$$t(m) + p(m)$$

the revenue from selling **the good** to consumer and **the data** to third party

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Our results extend to payoffs $v(\theta, a)$ such that:

- The third party's optimal action is a strictly increasing and concave function of posterior mean
- The third party's indirect value of a record is a positive and strictly convex function of posterior mean

Payoffs: Consumer

Type-θ consumer's payoff from reporting m when third party's action is a:

$$x(m)\Big(\theta - a \cdot \ell(\theta)\Big) - t(m),$$

$\ell(\theta)$ is type-θ's marginal disutility from data use $\rightsquigarrow$ **privacy loss** (1-dim type)

Instrumental (as opposed to intrinsic) preference for privacy

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For this talk, I'll focus on **affine** and **positive** privacy loss function

$$\boxed{\ell(\theta) = \alpha + \beta\theta \geq 0}$$

I refer to the slope parameter as the "**privacy β**"

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discussion

Discussion

Important assumptions:

- One-dimensional consumer type θ

  Type θ jointly affects the consumer's valuation for good, her privacy loss, and how valuable she is to third party

- Analysis focuses on deterministic mechanisms

Less important assumptions:

- All-or-nothing disclosure, disclosure of records is verifiable
- Functional forms

the economics of third-party data sharing

Outcomes

Let an *outcome* (or direct mechanism) be described by (x, t, a) where

- $x : \Theta \to \{0, 1\}$ is the allocation
- $t : \Theta \to \mathbb{R}$ is the payment
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An outcome (x, t, a) is

- *individually rational* (IR) if $U(\theta) \geq 0$ for all θ
- *incentive compatible* (IC) if $U(\theta) \geq u(\theta, \theta')$ for all θ, θ'
- *obedient* (OB) if $\tilde{a} \in \arg\max_{a'} \mathbb{E}\big(v(\theta, a') \mid a(\theta) = \tilde{a}\big)$ for all $\tilde{a} > 0$

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An outcome is *implementable* in a policy regime if it can be induced by an equilibrium for that regime

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Remark 1

An outcome (x, t, a) is implementable under

1. Market iff it is IR+IC+OB
2. Anarchy iff IR+IC+OB, and every buyer's data is sold $x(\theta) = 1 \Rightarrow a(\theta) > 0$
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1. Seller can commit to data-pricing strategy $\Rightarrow$ Standard application of RP

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3. Ban iff it is IR+IC+OB, and no data is sold $(a \equiv 0)$

2. Seller can't commit to data-pricing strategy: Each record has positive value for TP $\Rightarrow$ it will be sold $\Rightarrow a(\theta) > 0$ if $x(\theta) = 1$

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3. Seller can't sell records $\Rightarrow a \equiv 0$ (OB is vacuous)

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Implication: Seller's revenue is weakly higher in a data market

How Privacy β Affects Consumer Rents

Question: Does a policy leading to an increase in data use (i.e., a pointwise increase in $a : \Theta \to \mathbb{R}$) benefit or hurt consumers, keeping x fixed?

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Recall consumers dislike data use by assumption (since $\ell(\theta) \geq 0$)

Yet, *in equilibrium*, more data use is not necessarily bad for consumers, since transfers adjust when data use changes

The question is whether the transfer adjustment compensates consumers for their privacy loss

How Privacy β Affects Consumer Rents

The answer reveals the importance of the privacy β

Proposition 1

Among implementable outcomes with the same allocation rule x:

- If $\beta \geq 0,$ an increase in data use **decreases** consumer payoff $U(\theta)$ for all θ
- If $\beta \leq 0,$ an increase in data use **increases** consumer payoff $U(\theta)$ for all θ

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Economic Implication:

- $\sim$ Optimal policy should discourage data use when $\beta > 0$
- $\sim$ Optimal policy should encourage data use when $\beta \leq 0$

How Privacy β Affects Consumer Rents

Intuition: Consider the following thought experiment

- Fix an allocation x and suppose data use increases uniformly
- Seller will then lower t (uniformly) to keep marginal purchaser indifferent
- Inframarginal purchaser benefits from price reduction (lower t) but suffers a privacy loss (higher a)
- When $\beta > 0$, higher types suffer more from privacy loss than the marginal type and are therefore undercompensated
- When $\beta < 0$, higher types suffer less and thus are overcompensated

How Privacy β Shapes Data Sharing

So far, we have held the allocation rule x fixed

The seller's incentive to serve additional consumers depends on how their data can be monetized

To understand this, we next characterize what kind of data sharing is feasible

Here, too, the sign of the privacy β plays a central role

How Privacy β Shapes Data Sharing

Proposition 2

Consider any policy regime in which data sales are permitted:

- If $\beta > 0$, third-party data sharing is "**coarse**"

  In any implementable outcome, at most one data record is sold; this record pools all types of consumers whose data is sold

- If $\beta \leq 0$, third-party data sharing can be "**granular**"

  There is an implementable outcome in which a continuum of data records is sold, each revealing a unique consumer type

How Privacy β Shapes Data Sharing

Economic Implications:

- If $\beta > 0$

  Consumers whose data is sold must experience **the same** data utilization, and are compensated equally, even if the value of their data is different, and even if their WTP for privacy differs

- If $\beta < 0$

  Consumers whose data is sold can experience **different** data utilization, and thus can be compensated differently

How Privacy β Shapes Data Sharing

Intuition: If $\boxed{\beta > 0}$

- $\rightsquigarrow$ IC $\Rightarrow$ seller must ensure that $a(\theta)$ is **decreasing** on the set of trading types

  High types are more privacy sensitive, so they must not be assigned more data use to prevent deviations

- $\rightsquigarrow$ OB + monotone $a \Rightarrow$ seller must ensure that $a(\theta)$ is **increasing** on the set of trading types

  Higher types call for higher actions from third party

Thus, IC and OB are **in conflict** $\rightsquigarrow$ coarse disclosure

How Privacy β Shapes Data Sharing

Intuition: If $\beta < 0$

- $\sim$ IC $\Rightarrow$ seller must ensure that $a(\theta)$ is **increasing** on the set of trading types.

  Low types care more about privacy, so they must not be assigned more data use to prevent deviations

- $\sim$ OB + monotone $a \Rightarrow$ seller must ensure that $a(\theta)$ is **non-decreasing** on the set of trading types

  This is because higher types call for higher actions from third party

Thus, IC and OB are **not in conflict** $\rightsquigarrow$ granular disclosure is feasible

Taking Stock

We showed how the sign of the privacy β plays crucial role in shaping

1. **Consumer rents:** Whether more data use helps or hurts consumers
2. **Data sharing:** Whether data disclosure is coarse or granular

Thus the privacy β shapes equilibrium and optimal regulation

Next, we analyze separately the two cases: **positive** and **negative**

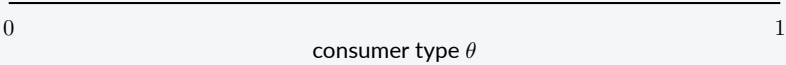
positive β

(higher types are more privacy-sensitive)

Equilibrium When $\beta > 0$

Proposition 3

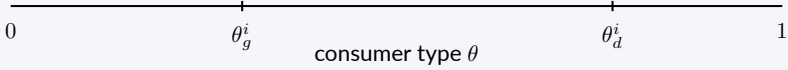
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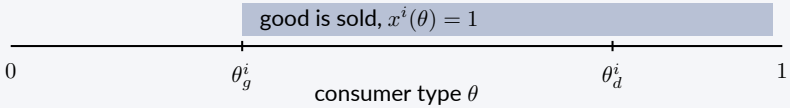
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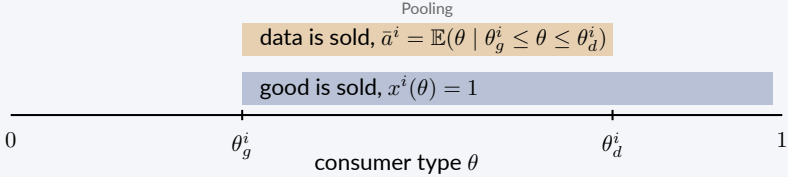
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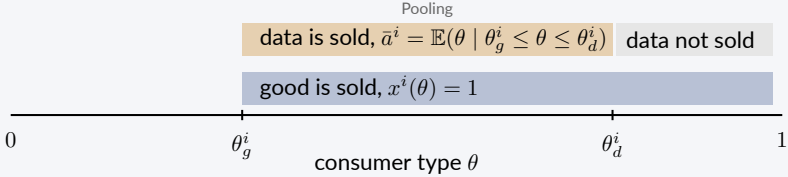
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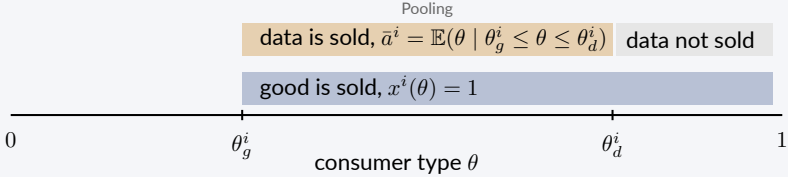
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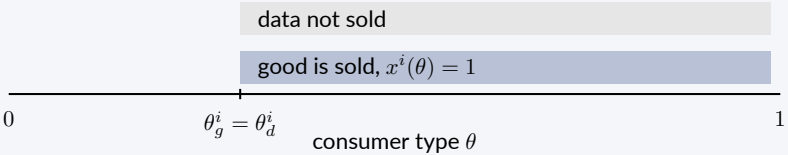
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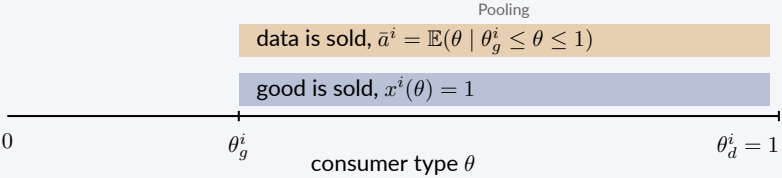
In particular, under **Data Ban**, $\theta_d^{\mathsf{B}} = \theta_g^{\mathsf{B}} \rightsquigarrow$ no data is sold

- No aftermarket $\Rightarrow$ standard monopoly pricing problem
- Simple posted price mechanism

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Fix $\beta > 0$ and a policy regime $i \in \{\mathsf{A}, \mathsf{B}, \mathsf{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that a seller-optimal outcome (x^i, a^i, t^i) is as follows:



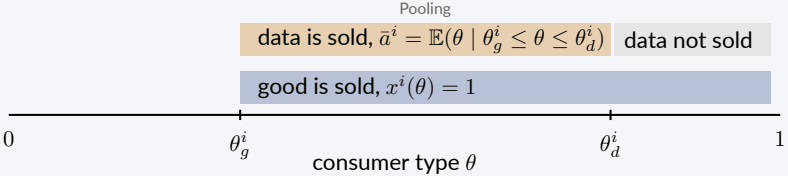
In particular, under **Data Anarchy**, $\theta_d^{\mathsf{A}} = 1 \rightsquigarrow$ all data is sold

- Monetization is coarse; consumers are pooled in same record
- Still posted price. Same treatment, same "compensation"

Equilibrium When $\beta > 0$

Proposition 3

Fix $\beta > 0$ and a policy regime $i \in \{\mathsf{A}, \mathsf{B}, \mathsf{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that a seller-optimal outcome (x^i, a^i, t^i) is as follows:



In particular, under **Data Market**, $\theta_d^{\mathsf{M}} \in [\theta_g^{\mathsf{M}}, 1] \rightsquigarrow$ some data is sold

- Two versions of the good are offered: one is private, the other is not
- Private version attracts higher types, who are more privacy sensitive

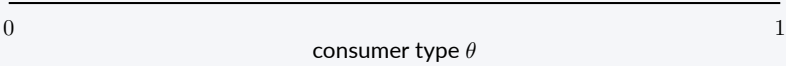
negative β

(higher types are less privacy-sensitive)

Equilibrium When $\beta \leq 0$

Proposition 4

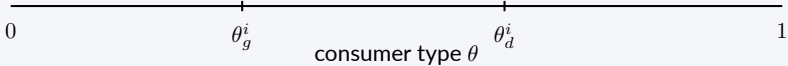
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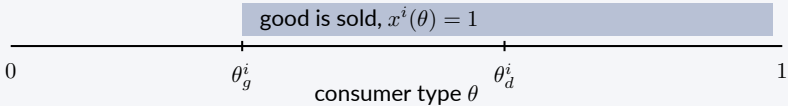
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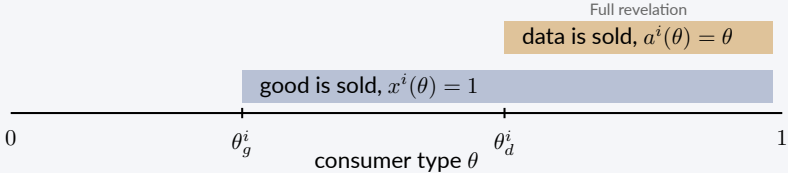
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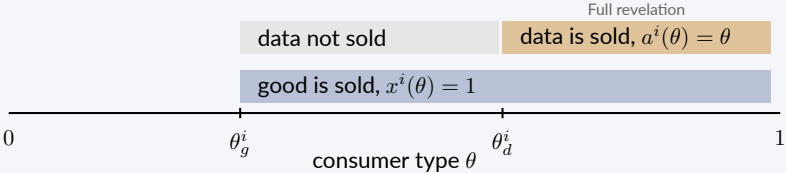
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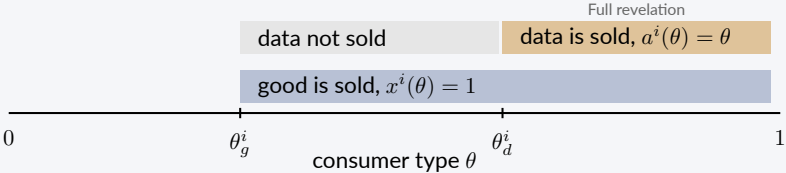
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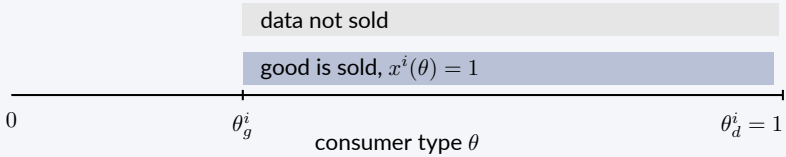
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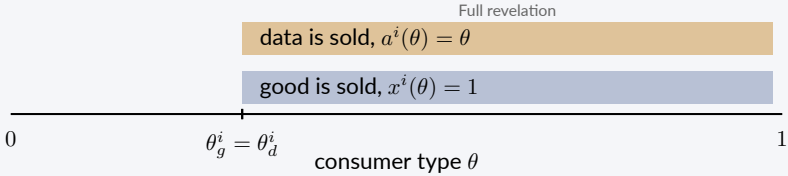
In particular, under **Data Ban**, $\theta_d^{\mathsf{B}} = 1 \rightsquigarrow$ no data is sold

- No aftermarket $\Rightarrow$ standard monopoly pricing problem
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Fix $\beta \leq 0$ and a policy regime $i \in \{\mathsf{A}, \mathsf{B}, \mathsf{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that an equilibrium outcome (x^i, a^i, t^i) is as follows:



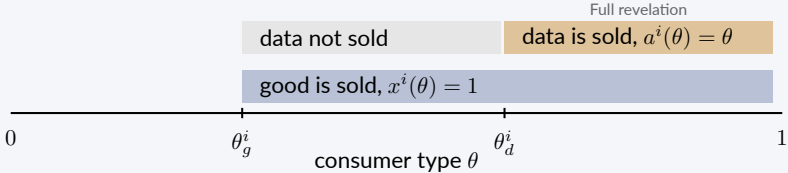
In particular, under **Data Anarchy**, $\theta_d^{\mathsf{A}} = \theta_g^{\mathsf{A}} \rightsquigarrow$ all data is sold

- Monetization is granular; third party learns θ (feasible vs optimal)
- Rich menu: each version with a different data use and compensation

Equilibrium When $\beta \leq 0$

Proposition 4

Fix $\beta \leq 0$ and a policy regime $i \in \{\mathsf{A}, \mathsf{B}, \mathsf{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that an equilibrium outcome (x^i, a^i, t^i) is as follows:



In particular, under **Data Market**, $\theta_d^{\mathsf{M}} \in [\theta_g^{\mathsf{M}}, 1] \rightsquigarrow$ some data is sold

- Many versions of the good are offered: one is private, the others are not
- Private version draws lower types, who are more privacy sensitive

welfare

How Policies Shape Welfare

What are the welfare implications of these policies?

Seller is weakly better off in Data Market (**Remark 1**)

Third party is indifferent across regimes

What about consumers?

How Policies Shape Welfare

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Seller is weakly better off in Data Market (**Remark 1**)

Third party is indifferent across regimes

What about consumers?

- ∼ **Policy Puzzle:** Without further structure, each policy regime can be best for all consumers

Welfare Implications

Proposition 5

If $\beta \leq 0$

$\sim$ Data Market Pareto dominates Data Ban

If $\beta > 0$

$\sim$ Data Market Pareto dominates Anarchy (when β sufficiently small)

Welfare Implications

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- $\sim$ Types below a cutoff $\tilde{\theta}$ prefer Market; those above prefer Anarchy

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Welfare Implications

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  Test: Market Pareto dominates Anarchy iff $p^{\mathsf{M}}_{\min} \leq p^{\mathsf{A}}_{\min}$ (compare prices of cheapest versions)

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If $\beta > 0$

- $\sim$ Data Market Pareto dominates Anarchy (when β sufficiently small)
- $\sim$ Types below a cutoff $\hat{\theta}$ prefer Market; those above prefer Ban

  Test: Market Pareto dominates Ban iff $p^{\mathsf{M}}_{\max} \leq p^{\mathsf{B}}$ (compare prices of private versions)

Summary

Therefore, Data Market has many desirable welfare properties

1. It is never Pareto dominated
2. It Pareto dominates either data anarchy or a data ban—with the sign of the privacy β determining which
3. When it is not Pareto dominant, it protects lower-valuation consumers

In this sense, establishing a data market emerges as an attractive policy choice, performing well across environments

conclusion

Conclusion

A central concern of privacy law is how to regulate third-party data sharing

A tractable framework to study third-party data sharing and optimal regulation

Positive Results:

- ∼ We identify a key structural parameter: the privacy β

  It shapes data monetization, consumer rents, and welfare rankings across policies

Normative Results:

- ∼ No single policy is best for all consumers in every environment
- ∼ Yet, **Data Markets** emerge as a robust, though imperfect, compromise

Privacy For Sale

Piotr Dworczak
Cologne, Northwestern

Jacopo Perego
Columbia

Laura Veldkamp
Columbia

Thanks!